

Router registration audit (source of truth) + second-pass validation

- FastAPI app mounts one top router: `app.include_router(router)`
`/home/runner/work/CSCRS/CSCRS/api/app.py:110`
 - Aggregation happens in `api/routes.py` with explicit `include_router` calls:
`/home/runner/work/CSCRS/CSCRS/api/routes.py:71-87`
 - Second-pass check for missed routers: only these `include_router` calls exist in repo (rg `include_router`), no hidden extra API mounts.
 - Total operations found from decorators: **96**
(GET 45, POST 34, PATCH 16, DELETE 1) across `api/*.py`.
-

2) Operations count by logical group

- Health 3
 - Auth 13
 - Department Admin 2
 - City Admin 8
 - Worker 6
 - Department 5
 - Assignment 3
 - Report 10
 - Timeline 1
 - Resolution 5
 - Forward Request 8
 - Dashboard 14
 - Notifications 4
 - Profile 4
 - Feedback 2
 - System Issue 5
 - AI Dataset 1
 - Generated PDF reports 2
- Total = 96**
-

3) Complete endpoint inventory

(method|full_path|group|roles|request_type|response_type|source_file)

m e t h o d	full_path	gr ou p	roles	request_t ype	response _type	source_file
G E T	/health	He alt h	Public	none	dict	/home/runner/w ork/CSCRS/CSC RS/api/routes.py :27
G E T	/liveness	He alt h	Public	none	dict	/home/runner/w ork/CSCRS/CSC RS/api/routes.py :38
G E T	/readiness	He alt h	Public	none	none/503	/home/runner/w ork/CSCRS/CSC RS/api/routes.py :44
P O S T	/api/v1/auth /register	Au th	Public	UserCrea te JSON	MessageR esponse	/home/runner/w ork/CSCRS/CSC RS/api/auth.py:4 3
P O S T	/api/v1/auth /login	Au th	Public	OAuth2Pa sswordRe questFor m	TokenRes ponse	/home/runner/w ork/CSCRS/CSC RS/api/auth.py:6 7
G E T	/api/v1/auth /me	Au th	Authentica ted	bearer only	dict user	/home/runner/w ork/CSCRS/CSC RS/api/auth.py:9 5
P O S T	/api/v1/auth /refresh	Au th	Public (refresh token body)	RefreshTo kenReque st	TokenRes ponse	/home/runner/w ork/CSCRS/CSC RS/api/auth.py:1 05

m e t h o d	full_path	gr o u p	roles	request_t ype	response _type	source_file
P O S T	/api/v1/auth /logout	Au th	Public (refresh token body)	RefreshTo kenReque st	dict	/home/runner/w ork/CSCRS/CSC RS/api/auth.py:1 19
P O S T	/api/v1/auth /logout-all	Au th	Authentica ted	none	dict	/home/runner/w ork/CSCRS/CSC RS/api/auth.py:1 32
G E T	/api/v1/auth /sessions	Au th	Authentica ted	none	list[Sessi onRespon se]	/home/runner/w ork/CSCRS/CSC RS/api/auth.py:1 45
P O S T	/api/v1/auth /verify- email	Au th	Public	VerifyEma ilRequest	MessageR esponse	/home/runner/w ork/CSCRS/CSC RS/api/auth.py:1 59
P O S T	/api/v1/auth /resend-otp	Au th	Public	ResendO TPReques t	MessageR esponse	/home/runner/w ork/CSCRS/CSC RS/api/auth.py:1 82
P O S T	/api/v1/auth /forgot- password	Au th	Public	ForgotPas swordReq uest	MessageR esponse	/home/runner/w ork/CSCRS/CSC RS/api/auth.py:2 03
P O S T	/api/v1/auth /verify- reset-otp	Au th	Public	VerifyRes etOTPReq uest	MessageR esponse	/home/runner/w ork/CSCRS/CSC RS/api/auth.py:2 24
P O	/api/v1/auth /reset- password	Au th	Public	ResetPas swordReq uest	MessageR esponse	/home/runner/w ork/CSCRS/CSC

m e t h o d	full_path	gr o u p	roles	request_t ype	response _type	source_file
S T						RS/api/auth.py:246
P O S T	/api/v1/auth/change-password	Auth	Authenticated	ChangePasswordRequest	MessageResponse	/home/runner/work/CSCRS/CSCRS/api/auth.py:269
P O S T	/api/v1/admins	DepartmentAdmin	CityAdmin/SuperAdmin	DepartmentAdminCreate	DepartmentAdminResponse	/home/runner/work/CSCRS/CSCRS/api/admin.py:27
P O S T	/api/v1/admins/activate	DepartmentAdmin	Public token flow	DepartmentAdminActivationRequest	DepartmentAdminActivationResponse	/home/runner/work/CSCRS/CSCRS/api/admin.py:54
P O S T	/api/v1/city-admins	CityAdmin	SuperAdmin	CityAdminCreate	CityAdminResponse	/home/runner/work/CSCRS/CSCRS/api/city_admin.py:29
P O S T	/api/v1/city-admins/activate	CityAdmin	Public token flow	CityAdminActivationRequest	CityAdminActivationResponse	/home/runner/work/CSCRS/CSCRS/api/city_admin.py:56

m e t h o d	full_path	gr ou p	roles	request_t ype	response _type	source_file
P A T C H	/api/v1/city-admins/citizens/{citizen_id}/block	Cit y Ad mi n	CityAdmin	path+BlockCitizenRequest	dict	/home/runner/work/CSCRS/CSCRS/api/city_admin.py:80
P A T C H	/api/v1/city-admins/citizens/{citizen_id}/unblock	Cit y Ad mi n	CityAdmin	path	dict	/home/runner/work/CSCRS/CSCRS/api/city_admin.py:99
P A T C H	/api/v1/city-admins/departments-admins/{admin_id}/block	Cit y Ad mi n	CityAdmin	path+BlockDepartmentAdminRequest	dict	/home/runner/work/CSCRS/CSCRS/api/city_admin.py:114
P A T C H	/api/v1/city-admins/departments-admins/{admin_id}/unblock	Cit y Ad mi n	CityAdmin	path	dict	/home/runner/work/CSCRS/CSCRS/api/city_admin.py:132
P A T C H	/api/v1/city-admins/admins/{admin_id}/block	Cit y Ad mi n	SuperAdmin	path+BlockCityAdminRequest	dict	/home/runner/work/CSCRS/CSCRS/api/city_admin.py:146
P A T C H	/api/v1/city-admins/admins/{admin_id}/unblock	Cit y Ad mi n	SuperAdmin	path	dict	/home/runner/work/CSCRS/CSCRS/api/city_admin.py:165

m e t h o d	full_path	gr o u p	roles	request_t ype	response _type	source_file
P O S T	/api/v1/workers	W o r k e r	Departme ntAdmin/S uperAdmi n	WorkerCr eate	WorkerRe sponse	/home/runner/w ork/CSCRS/CSC RS/api/worker.py :27
P O S T	/api/v1/workers/activate	W o r k e r	Public token flow	WorkerAc tivationRe quest	WorkerAc tivationRe sponse	/home/runner/w ork/CSCRS/CSC RS/api/worker.py :65
P A T C H	/api/v1/workers/{worker_id}/deactivate	W o r k e r	Departme ntAdmin/S uperAdmi n	path	WorkerSt atusResp onse	/home/runner/w ork/CSCRS/CSC RS/api/worker.py :84
P A T C H	/api/v1/workers/{worker_id}/activate	W o r k e r	Departme ntAdmin/S uperAdmi n	path	WorkerSt atusResp onse	/home/runner/w ork/CSCRS/CSC RS/api/worker.py :107
P A T C H	/api/v1/workers/{worker_id}/block	W o r k e r	Departme ntAdmin	path+Blo ckWorker Request	WorkerSt atusResp onse	/home/runner/w ork/CSCRS/CSC RS/api/worker.py :128
P A T C H	/api/v1/workers/{worker_id}/unblock	W o r k e r	Departme ntAdmin	path	WorkerSt atusResp onse	/home/runner/w ork/CSCRS/CSC RS/api/worker.py :151
P O S T	/api/v1/departments	De pa rt	SuperAdm in	Departme ntCreate	Departme ntRespon se	/home/runner/w ork/CSCRS/CSC RS/api/departme nt.py:30

m e t h o d	full_path	gr o u p	roles	request_t ype	response _type	source_file
		me nt				
G E T	/api/v1/departments	De pa rt me nt	SuperAdmin/CityAdmin	none	list[DepartmentResponse]	/home/runner/work/CSCRS/CSCRS/api/department.py:59
G E T	/api/v1/departments/{department_id}	De pa rt me nt	SuperAdmin/CityAdmin	path	DepartmentResponse	/home/runner/work/CSCRS/CSCRS/api/department.py:76
P A T C H	/api/v1/departments/{department_id}/activate	De pa rt me nt	SuperAdmin/CityAdmin	path	DepartmentStatusResponse	/home/runner/work/CSCRS/CSCRS/api/department.py:105
P A T C H	/api/v1/departments/{department_id}/deactivate	De pa rt me nt	SuperAdmin/CityAdmin	path	DepartmentStatusResponse	/home/runner/work/CSCRS/CSCRS/api/department.py:134
P O S T	/api/v1/assignments	As sig nm ent	DepartmentAdmin/SupperAdmin	AssignmentCreate	AssignmentResponse	/home/runner/work/CSCRS/CSCRS/api/assignment.py:24
G E T	/api/v1/assignments/my	As sig nm ent	Worker	none	list[WorkerAssignmentResponse]	/home/runner/work/CSCRS/CSCRS/api/assignment.py:46

m e t h o d	full_path	gr ou p	roles	request_t ype	response _type	source_file
P O S T	/api/v1/assi gnments/{a ssignment_i d}/start	As sig nm ent	Worker	path+Star tWorkReq uest	StartWork Response	/home/runner/w ork/CSCRS/CSC RS/api/assignme nt.py:63
P O S T	/api/v1/repo rt	Re po rt	Citizen	multipart file,descri ption	dict (duplicate /new- report payload)	/home/runner/w ork/CSCRS/CSC RS/api/report.py: 67
G E T	/api/v1/repo rts/my	Re po rt	Citizen	none	list[Citize nReportLi stItem]	/home/runner/w ork/CSCRS/CSC RS/api/report.py: 425
G E T	/api/v1/repo rts	Re po rt	CityAdmin	query page/filte r	Paginated CityRepor ts	/home/runner/w ork/CSCRS/CSC RS/api/report.py: 442
G E T	/api/v1/repo rts/departm ent	Re po rt	Departme ntAdmin	query status/pri ority/issu e_type	list[Depar tmentRep ortListIte m]	/home/runner/w ork/CSCRS/CSC RS/api/report.py: 481
G E T	/api/v1/repo rts/my/sear ch	Re po rt	Citizen	query que ry	list[Citize nReportLi stItem]	/home/runner/w ork/CSCRS/CSC RS/api/report.py: 506
G E T	/api/v1/repo rts/search	Re po rt	CityAdmin	query que ry	list[CityR eportListIte m]	/home/runner/w ork/CSCRS/CSC RS/api/report.py: 525

m e t h o d	full_path	gr ou p	roles	request_t ype	response _type	source_file
G E T	/api/v1/repo rts/departm ent/search	Re po rt	Departme ntAdmin	query que ry	list[Depar tmentRep ortListIte m]	/home/runner/w ork/CSCRS/CSC RS/api/report.py: 543
G E T	/api/v1/repo rts/{report_ number}	Re po rt	Citizen	path	ReportRe sponse	/home/runner/w ork/CSCRS/CSC RS/api/report.py: 562
P O S T	/api/v1/repo rts/{report_ id}/cancel	Re po rt	Departme ntAdmin	path+Rep ortCancel lationReq uest	dict	/home/runner/w ork/CSCRS/CSC RS/api/report.py: 589
P O S T	/api/v1/repo rts/{report_ id}/reopen	Re po rt	Departme ntAdmin	path+Rep ortReope nRequest	dict	/home/runner/w ork/CSCRS/CSC RS/api/report.py: 608
G E T	/api/v1/repo rts/{report_ id}/timeline	Ti me lin e	Citizen	path	TimelineR esponse	/home/runner/w ork/CSCRS/CSC RS/api/timeline. py:22
P O S T	/api/v1/reso lutions	Re sol uti on	Worker	multipart assignme nt_id,rem arks,imag e	Resolutio nRespons e	/home/runner/w ork/CSCRS/CSC RS/api/resolutio n.py:36
G E T	/api/v1/reso lutions/man ual-review	Re sol uti on	Departme ntAdmin	none	list[Manu alReviewI tem]	/home/runner/w ork/CSCRS/CSC RS/api/resolutio n.py:71

m e t h o d	full_path	gr o u p	roles	request_t ype	response _type	source_file
G E T	/api/v1/resolutions/manual-review/{report_id}	Resolution	DepartmentAdmin	path	ManualReviewDetail	/home/runner/work/CSCRS/CSCRS/api/resolution.py:89
P O S T	/api/v1/resolutions/manual-review/{report_id}/approve	Resolution	DepartmentAdmin	path	ResolutionResponse	/home/runner/work/CSCRS/CSCRS/api/resolution.py:112
P O S T	/api/v1/resolutions/manual-review/{report_id}/reject	Resolution	DepartmentAdmin	path+ManualReviewRejectRequest	ResolutionResponse	/home/runner/work/CSCRS/CSCRS/api/resolution.py:134
P O S T	/api/v1/forward-requests/{report_id}	Forward	Worker	path+ForwardRequestCreate	ForwardRequestResponse	/home/runner/work/CSCRS/CSCRS/api/forward_request.py:39
G E T	/api/v1/forward-requests/pending	Forward	DepartmentAdmin	none	list[ForwardRequestResponse]	/home/runner/work/CSCRS/CSCRS/api/forward_request.py:61
G E T	/api/v1/forward-requests/incoming	Forward	DepartmentAdmin	none	list[IncomingForwardRequestResponse]	/home/runner/work/CSCRS/CSCRS/api/forward_request.py:81

m e t h o d	full_path	gr o u p	roles	request_t ype	response _type	source_file
G E T	/api/v1/forward-requests/{request_id}	Forward	DepartmentAdmin	path	ForwardRequestDetailResponse	/home/runner/work/CSCRS/CSCRS/api/forward_request.py:101
P O S T	/api/v1/forward-requests/{request_id}/approve	Forward	DepartmentAdmin	path+ForwardRequestApprove	ForwardRequestResponse	/home/runner/work/CSCRS/CSCRS/api/forward_request.py:124
P O S T	/api/v1/forward-requests/{request_id}/accept	Forward	DepartmentAdmin	path	ForwardRequestResponse	/home/runner/work/CSCRS/CSCRS/api/forward_request.py:148
P O S T	/api/v1/forward-requests/{request_id}/decline	Forward	DepartmentAdmin	path+ForwardRequestReject	ForwardRequestResponse	/home/runner/work/CSCRS/CSCRS/api/forward_request.py:170
P O S T	/api/v1/forward-requests/{request_id}/reject	Forward	DepartmentAdmin	path+ForwardRequestReject	ForwardRequestResponse	/home/runner/work/CSCRS/CSCRS/api/forward_request.py:194
G E T	/api/v1/dashboard/summary	Dashboard	SuperAdmin/CityAdmin/DepartmentAdmin	none	CityDashboardSummaryResponse or department summary variant	/home/runner/work/CSCRS/CSCRS/api/dashboard.py:39

m e t h o d	full_path	gr ou p	roles	request_t ype	response _type	source_file
G E T	/api/v1/dashboard/feedback	Da sh bo ard	CityAdmin	none	Feedback Dashboar dResponse	/home/runner/work/CSCRS/CSCRS/api/dashboard.py:67
G E T	/api/v1/dashboard/departments	Da sh bo ard	CityAdmin	none	list[Depart mentStat isticsItem]	/home/runner/work/CSCRS/CSCRS/api/dashboard.py:82
G E T	/api/v1/dashboard/issues	Da sh bo ard	CityAdmin	none	list[Issue StatisticsItem]	/home/runner/work/CSCRS/CSCRS/api/dashboard.py:96
G E T	/api/v1/dashboard/status	Da sh bo ard	CityAdmin	none	list[Status StatisticsItem]	/home/runner/work/CSCRS/CSCRS/api/dashboard.py:110
G E T	/api/v1/dashboard/priorities	Da sh bo ard	CityAdmin	none	list[Priorit yStatistic sItem]	/home/runner/work/CSCRS/CSCRS/api/dashboard.py:124
G E T	/api/v1/dashboard/monthly-trends	Da sh bo ard	CityAdmin	query yea r	list[Month lyTrendItem]	/home/runner/work/CSCRS/CSCRS/api/dashboard.py:138
G E T	/api/v1/dashboard/recent-reports	Da sh bo ard	CityAdmin	query limi t	list[Recen tReportItem]	/home/runner/work/CSCRS/CSCRS/api/dashboard.py:157
G E T	/api/v1/dashboard/high-priority	Da sh	CityAdmin	query limi t	list[HighP riorityRep ortItem]	/home/runner/work/CSCRS/CSC

m e t h o d	full_path	gr ou p	roles	request_t ype	response _type	source_file
		bo ard				RS/api/dashboar d.py:176
G E T	/api/v1/dash board/insig hts	Da sh bo ard	CityAdmin	none	list[Dashb oardInsig htItem]	/home/runner/w ork/CSCRS/CSC RS/api/dashboar d.py:195
G E T	/api/v1/dash board/depar tment/dash board	Da sh bo ard	Departme ntAdmin	none	Departme ntDashbo ardRespo nse	/home/runner/w ork/CSCRS/CSC RS/api/dashboar d.py:211
G E T	/api/v1/dash board/top- workers	Da sh bo ard	CityAdmin	query dep artment_i d,limit	list[TopW orkerItem]	/home/runner/w ork/CSCRS/CSC RS/api/dashboar d.py:230
G E T	/api/v1/dash board/work er/dashboar d	Da sh bo ard	Worker	none	WorkerDa shboardR esponse	/home/runner/w ork/CSCRS/CSC RS/api/dashboar d.py:249
G E T	/api/v1/dash board/citize n/dashboar d	Da sh bo ard	Citizen	none	CitizenDa shboardR esponse	/home/runner/w ork/CSCRS/CSC RS/api/dashboar d.py:266
G E T	/api/v1/notif ications	No tifi cat ion s	Authentica ted	none	list[Notifi cationRes ponse]	/home/runner/w ork/CSCRS/CSC RS/api/in_app_n otification.py:28
G E T	/api/v1/notif ications/unr ead-count	No tifi cat	Authentica ted	none	UnreadN otification Response	/home/runner/w ork/CSCRS/CSC RS/api/in_app_n otification.py:48

m e t h o d	full_path	gr o u p	roles	request_t ype	response _type	source_file
		ion s				
P A T C H	/api/v1/notif ications/{no tification_id }/read	No tifi cat ion s	Authentica ted	path	dict	/home/runner/w ork/CSCRS/CSC RS/api/in_app_n otification.py:71
P A T C H	/api/v1/notif ications/rea d-all	No tifi cat ion s	Authentica ted	none	dict	/home/runner/w ork/CSCRS/CSC RS/api/in_app_n otification.py:10 6
G E T	/api/v1/profi le/me	Pr ofil e	Authentica ted	none	ProfileRes ponse	/home/runner/w ork/CSCRS/CSC RS/api/profile.py :42
P A T C H	/api/v1/profi le/me	Pr ofil e	Authentica ted	UpdatePr ofileRequ est	MessageR esponse	/home/runner/w ork/CSCRS/CSC RS/api/profile.py :64
P O S T	/api/v1/profi le/photo	Pr ofil e	Authentica ted	multipart photo	dict	/home/runner/w ork/CSCRS/CSC RS/api/profile.py :87
D E L E T E	/api/v1/profi le/photo	Pr ofil e	Authentica ted	none	MessageR esponse	/home/runner/w ork/CSCRS/CSC RS/api/profile.py :108

m e t h o d	full_path	gr o u p	roles	request_t ype	response _type	source_file
P O S T	/api/v1/feed back	Fe ed ba ck	Authentica ted	Feedback Create	Feedback Response	/home/runner/w ork/CSCRS/CSC RS/api/feedback .py:27
G E T	/api/v1/feed back/export	Fe ed ba ck	SuperAdm in/CityAd min/Depar tmentAdm in (inline check)	query for mat	file stream	/home/runner/w ork/CSCRS/CSC RS/api/feedback .py:48
P O S T	/api/v1/issu es	Sy ste m lss ue	Authentica ted	multipart form + optional file attach ments	Systemlss ueRespon se	/home/runner/w ork/CSCRS/CSC RS/api/system_i ssue.py:39
G E T	/api/v1/issu es	Sy ste m lss ue	SuperAdm in/CityAd min (inline check)	query status/cat egory/rep orter/sear ch	list[Syste mlssueLis tItem]	/home/runner/w ork/CSCRS/CSC RS/api/system_i ssue.py:83
G E T	/api/v1/issu es/export	Sy ste m lss ue	SuperAdm in (inline check)	query for mat csv/e xcel	FileRespo nse	/home/runner/w ork/CSCRS/CSC RS/api/system_i ssue.py:124
G E T	/api/v1/issu es/{issue_n umber}	Sy ste m lss ue	SuperAdm in/CityAd min (inline check)	path	Systemlss ueDetailR esponse	/home/runner/w ork/CSCRS/CSC RS/api/system_i ssue.py:152

m e t h o d	full_path	gr o u p	roles	request_t ype	response _type	source_file
P A T C H	/api/v1/issu es/{issue_n umber}/stat us	Sy ste m Iss ue	SuperAdm in (inline check)	path+Syst emIssueS tatusUpd ate	MessageR esponse	/home/runner/w ork/CSCRS/CSC RS/api/system_i ssue.py:180
G E T	/api/v1/ai- dataset/exp ort	AI Da tas et	SuperAdm in (inline check)	query for mat	CSV/XLSX stream	/home/runner/w ork/CSCRS/CSC RS/api/ai_datase t.py:28
G E T	/api/v1/repo rts/departm ent/downlo ad	Ge ner ate d Re po rt	Departme ntAdmin	none	PDF stream	/home/runner/w ork/CSCRS/CSC RS/api/generate_ report.py:20
G E T	/api/v1/repo rts/city/dow nload	Ge ner ate d Re po rt	CityAdmin	none	PDF stream	/home/runner/w ork/CSCRS/CSC RS/api/generate_ report.py:52

4) Trace map (dependencies → schemas → services → CRUD/models → validators/side effects)

- **Auth chain:**
 OAuth2PasswordBearer + get_current_user (authentication/dependencies.py:20-77)
 → schemas/user.py request models/validators (:13-37, :74-130)
 → AuthService/RefreshTokenService (services/auth_service.py:42-741, services/refresh_token_service.py:28-237)
 → UserCRUD, EmailVerificationCRUD, PasswordResetCRUD, RefreshTokenCRUD, Login AuditCRUD →
 models users, email_verifications, password_resets, refresh_tokens, login_audits → side

effects: login audit rows, refresh-token rotation/revocation, email notifications (services/notification_service.py:102-208).

- **Report submission chain:**
require_citizen (api/report.py:79) → multipart/form params (api/report.py:77-80)
→ validate_uploaded_file (utils/file_utils.py:25-109) → AI inference + duplicate detection (services/duplicate_detection_service.py:30-112)
→ ReportService.create_report/save_image/save_detections (services/report_service.py:58-120) → CRUD report, report_image, report_detection, report_support → models reports, report_images, report_detections, report_supports → side effects: file writes/deletes, support increments, assignment auto-creation, audit log + in-app notifications (api/report.py:221-239, services/assignment.py:113-129).
- **Assignment chain:**
require_admin/require_worker (api/assignment.py:35,54,71)
→ AssignmentService (services/assignment.py:26-327)
→ AssignmentCRUD (database/crud/assignment.py) + calculate_distance check + status mutations (ReportStatus, AssignmentStatus) → side effects: audit logs + notifications.
- **Resolution chain:**
require_worker/require_department_admin (api/resolution.py:59,79-81) → form/file + validate_uploaded_file (services/resolution.py:451-470) → AI verification + rule engine (services/resolution.py:101-177)
→ ResolutionCRUD, ResolutionAttemptCRUD, ResolutionAIResultCRUD, AssignmentCRUD, report_crud → models resolutions, resolution_attempts, resolution_ai_results, assignments, reports → side effects: status transitions, notifications, email send (services/resolution.py:330-379, :540-777).
- **Forward request chain:**
worker/admin dependencies (api/forward_request.py)
→ ForwardRequestService (services/forward_request_service.py:31-825)
→ DepartmentForwardRequestCRUD, ReportService.forward_report, AssignmentService.assign_worker, UserCRUD, WorkerCRUD → models department_forward_requests, report_forward_history, assignments, reports → side effects: reassignment/cancel/revert flows, forward-count increments, notifications/audit logs.
- **Profile/system issue uploads:**
validate_uploaded_file reused for profile and issue attachments (services/profile_service.py:180-194, services/system_issue_service.py:74-96)
+ save_uploaded_file (utils/file_utils.py:118-159).

5) Authentication implementation summary

- Access token + refresh token JWT types (type: access|refresh) in authentication/security.py:73-81,191-199.
- Password hashing/verify via bcrypt/passlib (authentication/security.py:15-38).

- *****
uses OAuth2PasswordBearer(tokenUrl="/api/v1/auth/login") (authentication/dependencies.py:20-22).
- Login includes:
 - invalid credential auditing (services/auth_service.py:141-159, 216-234)
 - lockout after 5 failures for 30 min (services/auth_service.py:205-212)
 - inactive/blocked/email-not-verified gates (:242-259).
- Refresh rotates token and revokes previous (services/refresh_token_service.py:138-153).
- Logout single/all revokes refresh tokens (services/refresh_token_service.py:175-226).
- Session listing reads active refresh sessions (services/refresh_token_service.py:227-237).
- OTP/email verification:
 - registration generates OTP + email verification row (services/auth_service.py:84-97)
 - verify OTP with expiry/attempt limits (:380-414)
 - resend OTP replaces unverified records (:461-479)
- Password reset:
 - forgot password uses non-enumerating response (:502-512, 541-546)
 - verify reset OTP with attempts/expiry (:586-617)
 - reset password requires verified OTP (:662-676)
 - change password checks old/new/confirm (:704-729)
- Account activation for worker/department-admin/city-admin via invitation token (services/worker_invitation.py:35-82, services/worker_service.py:158-221, services/admin_service.py:122-169, services/city_admin_service.py:96-143).
- Blocking modeled on user (is_blocked, block_type, blocked_by, etc.) in database/models/user.py:48-73; enforced at auth dependency (authentication/dependencies.py:71-75).

6) RBAC summary (exact enum names + matrix + conditionals)

- Enum
members: UserRole.CITIZEN, WORKER, DEPARTMENT_ADMIN, CITY_ADMIN, SUPER_ADMIN (database/enums.py:4-11).
- Core dependency gates: require_roles(...) and role wrappers (authentication/dependencies.py:79-131).

- Conditional authorization examples:
 - Worker can start only own assignment (services/assignment.py:249-254)
 - Worker must be within radius (services/assignment.py:272-280)
 - Department-admin scope checks by department_id (worker mgmt, report cancel/reopen/forward, manual review) (services/worker_service.py:241-251, services/report_service.py:527-535, services/resolution.py:576-581)
 - Forward details limited to source/destination dept admins (services/forward_request_service.py:144-155)
 - Inline role sets on some endpoints using get_current_user:
 - feedback export (api/feedback.py:59-69)
 - dashboard summary (api/dashboard.py:47-57)
 - system issues list/detail/export/status (api/system_issue.py:102-110, 139-146, 166-174, 196-203)
 - ai dataset export super-admin only (api/ai_dataset.py:37-41).
-

7) Common API behavior

- Base version prefix mainly /api/v1 via include_router in aggregator (api/routes.py:72-87).
- Non-versioned health probes: /health, /liveness, /readiness (api/routes.py:27-66).
- Content types:
 - JSON (most)
 - application/x-www-form-urlencoded login (OAuth2PasswordRequestForm)
 - multipart/form-data uploads in report/resolution/system-issue/profile.
- Pagination: only /api/v1/reports with page, page_size, metadata (api/report.py:447-479, services/report_service.py:238-269).
- Filter/search:
 - reports filters by dept/status/priority/issue_type (database/crud/report.py:98-173)
 - search endpoints use ilike on report_number/issue_type (:174-235)
 - system issues supports status/category/reporter/search (database/crud/system_issue.py:54-120).
- IDs/report numbers:
 - Numeric IDs in most paths
 - report number format CSCRS-YYYYMMDD-<8HEX> (utils/report_number.py:5-11)

- issue number format ISS-YYYYMMDD-<8HEX> (utils/report_number.py:13-21).
 - Datetime handling:
 - UTC used heavily in token/audit/report flows (authentication/security.py, services/*).
 - CORS:
 - origins from env FRONTEND_BASE_URL split comma; allow_credentials true, methods/headers * (api/app.py:83-93).
 - Security headers middleware adds X-Content-Type-Options, X-Frame-Options, etc. (api/app.py:94-104).
 - Error handling:
 - custom RateLimitExceeded handler returns {success:false,message,error_code} 429 (api/app.py:56-79)
 - no custom RequestValidationError handler; FastAPI default 422 applies.
-

8) Upload audit

- Central validator checks:
 - filename present
 - extension allowlist
 - MIME allowlist
 - empty file rejection
 - size limit (413 if exceeded)
 - magic-byte/signature checks for jpg/jpeg/png/webp and mp4/mov/avi/mkv (utils/file_utils.py:25-109).
- Endpoint-specific limits/fields:
 - report submit: field file, image-only, MAX_FILE_SIZE (10MB) (api/report.py:77-104, configs/config.py:175)
 - resolution submit: field image, image-only, 10MB (services/resolution.py:451-470)
 - profile photo: field photo, image-only, 5MB (services/profile_service.py:180-194)
 - system issue: field attachments (single in API signature), max 5 attachments in service loop, images+videos, 10MB each (api/system_issue.py:61, services/system_issue_service.py:66-96).
- Storage:
 - UUID-based filenames (utils/file_utils.py:111-117)

- folders: uploads/, uploads/resolution, uploads/profile, uploads/system_issues.
 - Error behavior: mostly 400 for invalid type/signature/empty, 413 for oversize.
-

9) Rate limiting audit

- Backend: slowapi.Limiter with Redis storage REDIS_URL (utils/rate_limiter.py:75-79).
 - Keying: authenticated user (***** else client IP (utils/rate_limiter.py:59-73`)).
 - Limited endpoints/limits:
 - auth register 3 per minutes
 - login 5 per minute
 - verify-email 5 per minutes
 - resend-otp 5 per 10 minute
 - forgot-password 3 per 15 minutes
 - verify-reset-otp 5 per 10 minutes
 - reset-password 5 per 15 minutes
 - report submit 60 per hour
 - resolution upload 20 per hour
 - feedback submit 10 per hour
 - system issue submit 20 per hour
 - department report download 20 per hour
 - city report download 20 per hour
(all from decorator lines in api/*.py).
 - Response behavior: custom 429 JSON in app exception handler (api/app.py:56-79).
-

10) Error/status behavior findings

- Frequent HTTPException status patterns:
 - 400: validation/business rule (bad state, invalid OTP, bad format)
 - 401: auth token/credential/refresh invalid
 - 403: role/access/blocked/inactive/location
 - 404: entity missing
 - 409: conflicts (already assigned, duplicate, already blocked, etc.)
 - 413: upload size

- 423: account temporary lockout (services/auth_service.py:179-185)
 - 429: rate limit
 - 503: readiness DB disconnected (api/routes.py:60-66)
 - No global normalization besides 429 handler; responses are mixed typed models and ad-hoc dicts.
-

11) Major workflows + state transitions

- **Registration/verification:** Citizen created inactive/unverified → OTP verify sets is_active=true, is_email_verified=true (services/auth_service.py:70-83, 416-421).
 - **Login/session lifecycle:** login creates access+refresh + audit; refresh rotates; logout revokes token; logout-all revokes all sessions (services/auth_service.py:270-340, services/refresh_token_service.py:83-226).
 - **Password recovery:** forgot → reset OTP row → verify OTP → reset password (services/auth_service.py:492-695).
 - **Report lifecycle:**
 - create report PENDING
 - auto-assign -> ASSIGNED
 - start work -> IN_PROGRESS
 - resolution pass/manual review/reject path -> RESOLVED or stays/reverts
 - cancel -> CANCELLED
 - reopen -> back to PENDING + reassign (services/assignment.py, services/resolution.py, services/report_service.py).
 - **Forwarding workflow:** request PENDING → source approve WAITING_DESTINATION → destination accept ACCEPTED (department moved/reassign) or decline/reject REJECTED (revert assignment/report state) (services/forward_request_service.py).
 - **System issue workflow:** OPEN → IN_REVIEW/RESOLVED/REJECTED; closed_at set on terminal states (database/crud/system_issue.py:151-160).
 - **Blocking/unblocking:** user flags updated (UserCRUD.block_user/unblock_user) and enforced by auth dependency (authentication/dependencies.py:71-75).
-

12) Existing docs conflicts vs implementation

Key mismatches:

1. **Assignment start path**

- Docs: `/api/v1/assignments/{assignment_id}/start-work` (docs/API.md:45, docs/ARCHITECTURE.md:261)
- Code: `/api/v1/assignments/{assignment_id}/start` (api/assignment.py:63)

2. Resolution manual-review paths

- Docs use `/manual-reviews` and `/approve-manual-review` (docs/API.md:54-57, docs/ARCHITECTURE.md:354)
- Code uses `/manual-review/.../approve|reject` (api/resolution.py:71-135)

3. Forward request base path and create signature

- Docs: `/api/v1/forward-requests` body contains `report_id` (docs/API.md:65)
- Code: POST `/api/v1/forward-requests/{report_id}` (api/forward_request.py:34,39 + api/routes.py:82)

4. Dashboard base path

- Docs list `/dashboard/...` (docs/API.md:76-93)
- Actual includes `/api/v1/dashboard/...` (api/routes.py:79, api/dashboard.py:35)

5. Timeline/notifications/system-issues paths

- Docs: `/api/v1/timeline/...`, `/api/v1/in-app-notifications`, `/api/v1/system-issues` (docs/API.md:130-136)
- Code: `/api/v1/reports/{id}/timeline`, `/api/v1/notifications`, `/api/v1/issues...` (api/timeline.py:17-24, api/in_app_notification.py:23, api/system_issue.py:33)

13) Documentation gaps / ambiguities

- No single canonical endpoint list aligned to current router registration (docs are partially stale).
- Response envelope inconsistent (model vs plain dict vs file stream).
- Explicit status codes are sparse in decorators; most rely on defaults/exceptions.
- Some role restrictions happen inside handler body (inline `allowed_roles`), not only dependencies.
- Upload constraints and magic-byte checks are not centralized in docs though strict in code.
- Rate-limit key strategy (user-id fallback to IP) not documented.

14) Proposed docs/api-reference modular file tree

- docs/api-reference/00-overview.md
- docs/api-reference/01-routing-and-versioning.md

- docs/api-reference/02-authentication.md
- docs/api-reference/03-rbac-and-authorization.md
- docs/api-reference/04-health-and-platform.md
- docs/api-reference/05-reports.md
- docs/api-reference/06-assignments.md
- docs/api-reference/07-resolutions.md
- docs/api-reference/08-forward-requests.md
- docs/api-reference/09-admin-and-user-management.md
- docs/api-reference/10-dashboard.md
- docs/api-reference/11-feedback.md
- docs/api-reference/12-system-issues.md
- docs/api-reference/13-profile-and-notifications.md
- docs/api-reference/14-ai-dataset-and-generated-reports.md
- docs/api-reference/15-upload-rules.md
- docs/api-reference/16-rate-limits.md
- docs/api-reference/17-errors-and-status-codes.md
- docs/api-reference/18-workflows-and-state-machines.md
- docs/api-reference/19-openapi-and-examples.md
- docs/api-reference/appendix-endpoint-inventory.md

15) Recommended chapter generation order

1. Routing/versioning source-of-truth
2. Auth + token/session lifecycle
3. RBAC + conditional ownership/department checks
4. Core report lifecycle (report/assignment/resolution/forward)
5. Supporting modules (dashboard/feedback/issues/profile/notifications)
6. Upload validation + rate limits + error model
7. Workflow state machines
8. Conflict reconciliation against existing docs
9. Final appendix with endpoint inventory table and counts.

1) Registered API operations (re-discovered from app + include_router + decorator pass)

- **Total registered operations:** 96
- **GET:** 45
- **POST:** 34
- **PATCH:** 16
- **PUT:** 0
- **DELETE:** 1
- **Other methods:** 0

Second-pass checks (APIRouter, include_router, route decorators) found no additional registered operations beyond these 96.

2) Routing fix verification

- **Malformed paths:** No malformed syntax found; duplicate versioning is the structural issue.
 - **Accidental duplicate routes (same method + path):** None found.
 - **Conflicting method + path combos:** None found.
 - **Singular/plural inconsistencies that look accidental:** /api/v1/report vs many /api/v1/reports/*.
 - **Endpoints defined but not registered:** None found among api/*.py routers.
-

3) FINAL endpoint inventory (current implementation)

Method	Full Path	Logical Group	Authentication	Allowed Roles	Source File
GET	/health	Health	Public	N/A	/home/runner/work/CSCRS/CSCRS/api/routes.py
GET	/liveness	Health	Public	N/A	/home/runner/work/CSCRS/CSCRS/api/routes.py
GET	/readiness	Health	Public	N/A	/home/runner/work/CSCRS/CSCRS/api/routes.py

Method	Full Path	Logical Group	Authentication	Allowed Roles	Source File
POST	/api/v1/auth/register	Authentication	Public	N/A	/home/runner/work/CSCRS/CSCRS/api/auth.py
POST	/api/v1/auth/login	Authentication	Other credential mechanism (username/password)	Unauthenticated	/home/runner/work/CSCRS/CSCRS/api/auth.py
GET	/api/v1/auth/me	Authentication	***** token required	Any authenticated role	/home/runner/work/CSCRS/CSCRS/api/auth.py
POST	/api/v1/auth/refresh	Authentication	Refresh token credential required	Any valid refresh-token session	/home/runner/work/CSCRS/CSCRS/api/auth.py
POST	/api/v1/auth/logout	Authentication	Refresh token credential required	Any valid refresh-token session	/home/runner/work/CSCRS/CSCRS/api/auth.py
POST	/api/v1/auth/logout-all	Authentication	***** token required	Any authenticated role	/home/runner/work/CSCRS/CSCRS/api/auth.py
GET	/api/v1/auth/sessions	Authentication	***** token required	Any authenticated role	/home/runner/work/CSCRS/CSCRS/api/auth.py

Method	Full Path	Logical Group	Authentication	Allowed Roles	Source File
POST	/api/v1/auth/verify-email	Authentication	Other credential mechanism (email+ OTP)	Unauthenticated flow	/home/runner/work/CSCRS/CSCRS/api/auth.py
POST	/api/v1/auth/resend-otp	Authentication	Other credential mechanism (email OTP flow)	Unauthenticated flow	/home/runner/work/CSCRS/CSCRS/api/auth.py
POST	/api/v1/auth/forgot-password	Authentication	Other credential mechanism (email recovery flow)	Unauthenticated flow	/home/runner/work/CSCRS/CSCRS/api/auth.py
POST	/api/v1/auth/verify-reset-otp	Authentication	Other credential mechanism (reset OTP)	Unauthenticated flow	/home/runner/work/CSCRS/CSCRS/api/auth.py
POST	/api/v1/auth/reset-password	Authentication	Other credential mechanism (reset OTP +	Unauthenticated flow	/home/runner/work/CSCRS/CSCRS/api/auth.py

Method	Full Path	Logical Group	Authentication	Allowed Roles	Source File
			new password)		
POST	/api/v1/auth/change-password	Authentication	***** token required	Any authenticated role	/home/runner/work/CSCRS/CSCRS/api/auth.py
POST	/api/v1/workers	Workers	***** token required	DepartmentAdmin, SuperAdmin	/home/runner/work/CSCRS/CSCRS/api/worker.py
POST	/api/v1/workers/activate	Workers	Invitation/activation token flow	Token holder	/home/runner/work/CSCRS/CSCRS/api/worker.py
PATCH	/api/v1/workers/{worker_id}/deactivate	Workers	***** token required	DepartmentAdmin, SuperAdmin	/home/runner/work/CSCRS/CSCRS/api/worker.py
PATCH	/api/v1/workers/{worker_id}/activate	Workers	***** token required	DepartmentAdmin, SuperAdmin	/home/runner/work/CSCRS/CSCRS/api/worker.py
PATCH	/api/v1/workers/{worker_id}/block	Workers	***** token required	DepartmentAdmin	/home/runner/work/CSCRS/CSCRS/api/worker.py
PATCH	/api/v1/workers/{worker_id}/unblock	Workers	***** token required	DepartmentAdmin	/home/runner/work/CSCRS/CSCRS/api/worker.py

Method	Full Path	Logical Group	Authentication	Allowed Roles	Source File
POST	/api/v1/admins	Department Admin	***** token required	CityAdmin, SuperAdmin	/home/runner/work/CSCRS/CSCRS/api/admin.py
POST	/api/v1/admins/activate	Department Admin	Invitation/activation token flow	Token holder	/home/runner/work/CSCRS/CSCRS/api/admin.py
POST	/api/v1/city-admins	City Admin	***** token required	SuperAdmin	/home/runner/work/CSCRS/CSCRS/api/city_admin.py
POST	/api/v1/city-admins/activate	City Admin	Invitation/activation token flow	Token holder	/home/runner/work/CSCRS/CSCRS/api/city_admin.py
PATCH	/api/v1/city-admins/citizens/{citizen_id}/block	City Admin	***** token required	CityAdmin	/home/runner/work/CSCRS/CSCRS/api/city_admin.py
PATCH	/api/v1/city-admins/citizens/{citizen_id}/unblock	City Admin	***** token required	CityAdmin	/home/runner/work/CSCRS/CSCRS/api/city_admin.py
PATCH	/api/v1/city-admins/department-admins/{admin_id}/block	City Admin	***** token required	CityAdmin	/home/runner/work/CSCRS/CSCRS/api/city_admin.py
PATCH	/api/v1/city-admins/department-	City Admin	***** token required	CityAdmin	/home/runner/work/CSCRS/CSCRS/api/city_admin.py

Method	Full Path	Logical Group	Authentication	Allowed Roles	Source File
	admins/{admin_id}/unlock				
PATCH	/api/v1/city-admins/admins/{admin_id}/block	City Admin	***** token required	SuperAdmin	/home/runner/work/CSCRS/CSCRS/api/city_admin.py
PATCH	/api/v1/city-admins/admins/{admin_id}/unlock	City Admin	***** token required	SuperAdmin	/home/runner/work/CSCRS/CSCRS/api/city_admin.py
POST	/api/v1/assignments	Assignments	***** token required	DepartmentAdmin, SuperAdmin	/home/runner/work/CSCRS/CSCRS/api/assignment.py
GET	/api/v1/assignments/my	Assignments	***** token required	Worker	/home/runner/work/CSCRS/CSCRS/api/assignment.py
POST	/api/v1/assignments/{assignment_id}/start	Assignments	***** token required	Worker	/home/runner/work/CSCRS/CSCRS/api/assignment.py
POST	/api/v1/resolutions	Resolution	***** token required	Worker	/home/runner/work/CSCRS/CSCRS/api/resolution.py
GET	/api/v1/resolutions/manual-review	Resolution	***** token required	DepartmentAdmin	/home/runner/work/CSCRS/CSCRS/api/resolution.py
GET	/api/v1/resolutions/manual-review/{report_id}	Resolution	***** token required	DepartmentAdmin	/home/runner/work/CSCRS/CSCRS/api/resolution.py

Method	Full Path	Logical Group	Authentication	Allowed Roles	Source File
POST	/api/v1/resolutions/manual-review/{report_id}/approve	Resolution	***** token required	DepartmentAdmin	/home/runner/work/CSCRS/CSCRS/api/resolution.py
POST	/api/v1/resolutions/manual-review/{report_id}/reject	Resolution	***** token required	DepartmentAdmin	/home/runner/work/CSCRS/CSCRS/api/resolution.py
POST	/api/v1/departments	Departments	***** token required	SuperAdmin	/home/runner/work/CSCRS/CSCRS/api/department.py
GET	/api/v1/departments	Departments	***** token required	SuperAdmin, CityAdmin	/home/runner/work/CSCRS/CSCRS/api/department.py
GET	/api/v1/departments/{department_id}	Departments	***** token required	SuperAdmin, CityAdmin	/home/runner/work/CSCRS/CSCRS/api/department.py
PATCH	/api/v1/departments/{department_id}/activate	Departments	***** token required	SuperAdmin, CityAdmin	/home/runner/work/CSCRS/CSCRS/api/department.py
PATCH	/api/v1/departments/{department_id}/deactivate	Departments	***** token required	SuperAdmin, CityAdmin	/home/runner/work/CSCRS/CSCRS/api/department.py
POST	/api/v1/report	Reports (core)	***** token required	Citizen	/home/runner/work/CSCRS/CSCRS/api/report.py

Method	Full Path	Logical Group	Authentication	Allowed Roles	Source File
GET	/api/v1/reports/my	Reports (core)	***** token required	Citizen	/home/runner/work/CSCRS/CSCRS/api/report.py
GET	/api/v1/reports	Reports (core)	***** token required	CityAdmin	/home/runner/work/CSCRS/CSCRS/api/report.py
GET	/api/v1/reports/department	Reports (core)	***** token required	DepartmentAdmin	/home/runner/work/CSCRS/CSCRS/api/report.py
GET	/api/v1/reports/my/search	Reports (core)	***** token required	Citizen	/home/runner/work/CSCRS/CSCRS/api/report.py
GET	/api/v1/reports/search	Reports (core)	***** token required	CityAdmin	/home/runner/work/CSCRS/CSCRS/api/report.py
GET	/api/v1/reports/department/search	Reports (core)	***** token required	DepartmentAdmin	/home/runner/work/CSCRS/CSCRS/api/report.py
GET	/api/v1/reports/{report_number}	Reports (core)	***** token required	Citizen	/home/runner/work/CSCRS/CSCRS/api/report.py
POST	/api/v1/reports/{report_id}/cancel	Reports (core)	***** token required	DepartmentAdmin	/home/runner/work/CSCRS/CSCRS/api/report.py
POST	/api/v1/reports/{report_id}/reopen	Reports (core)	***** token required	DepartmentAdmin	/home/runner/work/CSCRS/CSCRS/api/report.py
GET	/api/v1/dashboard/summary	Dashboard	***** token required	SuperAdmin, CityAdmin,	/home/runner/work/CSCRS/CSCRS/api/dashboard.py

Method	Full Path	Logical Group	Authentication	Allowed Roles	Source File
		Analytics		DepartmentAdmin	
GET	/api/v1/dashboard/feedback	Dashboard Analytics	***** token required	CityAdmin	/home/runner/work/CSCRS/CSCRS/api/dashboard.py
GET	/api/v1/dashboard/departments	Dashboard Analytics	***** token required	CityAdmin	/home/runner/work/CSCRS/CSCRS/api/dashboard.py
GET	/api/v1/dashboard/issues	Dashboard Analytics	***** token required	CityAdmin	/home/runner/work/CSCRS/CSCRS/api/dashboard.py
GET	/api/v1/dashboard/status	Dashboard Analytics	***** token required	CityAdmin	/home/runner/work/CSCRS/CSCRS/api/dashboard.py
GET	/api/v1/dashboard/priorities	Dashboard Analytics	***** token required	CityAdmin	/home/runner/work/CSCRS/CSCRS/api/dashboard.py
GET	/api/v1/dashboard/monthly-trends	Dashboard Analytics	***** token required	CityAdmin	/home/runner/work/CSCRS/CSCRS/api/dashboard.py
GET	/api/v1/dashboard/recent-reports	Dashboard Analytics	***** token required	CityAdmin	/home/runner/work/CSCRS/CSCRS/api/dashboard.py

Method	Full Path	Logical Group	Authentication	Allowed Roles	Source File
GET	/api/v1/dashboard/high-priority	Dashboard Analytics	***** token required	CityAdmin	/home/runner/work/CSCRS/CSCRS/api/dashboard.py
GET	/api/v1/dashboard/insights	Dashboard Analytics	***** token required	CityAdmin	/home/runner/work/CSCRS/CSCRS/api/dashboard.py
GET	/api/v1/dashboard/department/dashboard	Dashboard Analytics	***** token required	DepartmentAdmin	/home/runner/work/CSCRS/CSCRS/api/dashboard.py
GET	/api/v1/dashboard/top-workers	Dashboard Analytics	***** token required	CityAdmin	/home/runner/work/CSCRS/CSCRS/api/dashboard.py
GET	/api/v1/dashboard/worker/dashboard	Dashboard Analytics	***** token required	Worker	/home/runner/work/CSCRS/CSCRS/api/dashboard.py
GET	/api/v1/dashboard/citizen/dashboard	Dashboard Analytics	***** token required	Citizen	/home/runner/work/CSCRS/CSCRS/api/dashboard.py
GET	/api/v1/reports/{report_id}/timeline	Citizen Timeline	***** token required	Citizen	/home/runner/work/CSCRS/CSCRS/api/timeline.py
GET	/api/v1/notifications	In-App Notifications	***** token required	Any authenticated role	/home/runner/work/CSCRS/CSCRS/api/in_app_notification.py

Method	Full Path	Logical Group	Authentication	Allowed Roles	Source File
GET	/api/v1/notifications/unread-count	In-App Notifications	***** token required	Any authenticated role	/home/runner/work/CSCRS/CSCRS/api/in_app_notification.py
PATCH	/api/v1/notifications/{notification_id}/read	In-App Notifications	***** token required	Any authenticated role	/home/runner/work/CSCRS/CSCRS/api/in_app_notification.py
PATCH	/api/v1/notifications/read-all	In-App Notifications	***** token required	Any authenticated role	/home/runner/work/CSCRS/CSCRS/api/in_app_notification.py
POST	/api/v1/forward-requests/{report_id}	Forward Requests	***** token required	Worker	/home/runner/work/CSCRS/CSCRS/api/forward_request.py
GET	/api/v1/forward-requests/pending	Forward Requests	***** token required	DepartmentAdmin	/home/runner/work/CSCRS/CSCRS/api/forward_request.py
GET	/api/v1/forward-requests/incoming	Forward Requests	***** token required	DepartmentAdmin	/home/runner/work/CSCRS/CSCRS/api/forward_request.py
GET	/api/v1/forward-requests/{request_id}	Forward Requests	***** token required	DepartmentAdmin	/home/runner/work/CSCRS/CSCRS/api/forward_request.py

Method	Full Path	Logical Group	Authentication	Allowed Roles	Source File
POST	/api/v1/forward-requests/{request_id}/approve	Forward Requests	***** token required	DepartmentAdmin	/home/runner/work/CSCRS/CSCRS/api/forward_request.py
POST	/api/v1/forward-requests/{request_id}/accept	Forward Requests	***** token required	DepartmentAdmin	/home/runner/work/CSCRS/CSCRS/api/forward_request.py
POST	/api/v1/forward-requests/{request_id}/decline	Forward Requests	***** token required	DepartmentAdmin	/home/runner/work/CSCRS/CSCRS/api/forward_request.py
POST	/api/v1/forward-requests/{request_id}/reject	Forward Requests	***** token required	DepartmentAdmin	/home/runner/work/CSCRS/CSCRS/api/forward_request.py
GET	/api/v1/profile/me	Profile	***** token required	Any authenticated role	/home/runner/work/CSCRS/CSCRS/api/profile.py
PATCH	/api/v1/profile/me	Profile	***** token required	Any authenticated role	/home/runner/work/CSCRS/CSCRS/api/profile.py
POST	/api/v1/profile/photo	Profile	***** token required	Any authenticated role	/home/runner/work/CSCRS/CSCRS/api/profile.py
DELETE	/api/v1/profile/photo	Profile	***** token required	Any authenticated role	/home/runner/work/CSCRS/CSCRS/api/profile.py

Method	Full Path	Logical Group	Authentication	Allowed Roles	Source File
POST	/api/v1/feedback	Feedback	***** token required	Any authenticated role	/home/runner/work/CSCRS/CSCRS/api/feedback.py
GET	/api/v1/feedback/export	Feedback	***** token required	SuperAdmin, CityAdmin, DepartmentAdmin	/home/runner/work/CSCRS/CSCRS/api/feedback.py
POST	/api/v1/issues	System Issues	***** token required	Any authenticated role	/home/runner/work/CSCRS/CSCRS/api/system_issue.py
GET	/api/v1/issues	System Issues	***** token required	SuperAdmin, CityAdmin	/home/runner/work/CSCRS/CSCRS/api/system_issue.py
GET	/api/v1/issues/export	System Issues	***** token required	SuperAdmin	/home/runner/work/CSCRS/CSCRS/api/system_issue.py
GET	/api/v1/issues/{issue_number}	System Issues	***** token required	SuperAdmin, CityAdmin	/home/runner/work/CSCRS/CSCRS/api/system_issue.py
PATCH	/api/v1/issues/{issue_number}/status	System Issues	***** token required	SuperAdmin	/home/runner/work/CSCRS/CSCRS/api/system_issue.py

Method	Full Path	Logical Group	Authentication	Allowed Roles	Source File
GET	/api/v1/ai-dataset/export	AI Dataset	***** token required	SuperAdmin	/home/runner/work/CSCRS/CSCRS/api/ai_dataset.py
GET	/api/v1/reports/department/download	Generated Reports	***** token required	DepartmentAdmin	/home/runner/work/CSCRS/CSCRS/api/generate_report.py
GET	/api/v1/reports/city/download	Generated Reports	***** token required	CityAdmin	/home/runner/work/CSCRS/CSCRS/api/generate_report.py

4) API groups and counts

Group	Count
Health	3
Authentication	13
Workers	6
Department Admin	2
City Admin	8
Assignments	3
Resolution	5
Departments	5
Reports (core)	10
Dashboard Analytics	14

Group	Count
Citizen Timeline	1
In-App Notifications	4
Forward Requests	8
Profile	4
Feedback	2
System Issues	5
AI Dataset	1
Generated Reports	2
Total	96

(Checks: group sum = 96 exactly)

5) Rate limits (exact decorator strings)

Method	Endpoint	Exact Limit String	Effective Key Strategy
POST	/api/v1/auth/register	3 per minutes	user:{sub} if valid ***** exists, else ip:{client_ip}
POST	/api/v1/auth/login	5 per minute	user:{sub} if valid ***** exists, else ip:{client_ip}
POST	/api/v1/auth/verify-email	5 per minutes	user:{sub} if valid ***** exists, else ip:{client_ip}
POST	/api/v1/auth/resend-otp	5 per 10 minute	user:{sub} if valid ***** exists, else ip:{client_ip}

Method	Endpoint	Exact Limit String	Effective Key Strategy
POST	/api/v1/auth/forgot-password	3 per 15 minutes	user:{sub} if valid ***** exists, else ip:{client_ip}
POST	/api/v1/auth/verify-reset-otp	5 per 10 minutes	user:{sub} if valid ***** exists, else ip:{client_ip}
POST	/api/v1/auth/reset-password	5 per 15 minutes	user:{sub} if valid ***** exists, else ip:{client_ip}
GET	/api/v1/reports/department/download	20 per hour	user:{sub} if valid ***** exists, else ip:{client_ip}
GET	/api/v1/reports/city/download	20 per hour	user:{sub} if valid ***** exists, else ip:{client_ip}
POST	/api/v1/feedback	10 per hour	user:{sub} if valid ***** exists, else ip:{client_ip}
POST	/api/v1/resolutions	20 per hour	user:{sub} if valid ***** exists, else ip:{client_ip}
POST	/api/v1/issues	20 per hour	user:{sub} if valid ***** exists, else ip:{client_ip}
POST	/api/v1/report	60 per hour	user:{sub} if valid ***** exists, else ip:{client_ip}

6) Major API contract characteristics (current source)

- **API version/base prefix:** versioned surface is mostly under /api/v1; health endpoints are root (/health, /liveness, /readiness).

- **Authentication mechanism:** OAuth2 bearer token (OAuth2PasswordBearer) + JWT access token validation; refresh token flow is separate request credential.
- **UserRole enum values:** Citizen, Worker, DepartmentAdmin, CityAdmin, SuperAdmin.
- **Upload content types:**
 - Images: image/jpeg, image/png, image/webp
 - System issue attachments also allow videos: video/mp4, video/quicktime, video/x-msvideo, video/x-matroska.
- **Central upload validation behavior:** validate_uploaded_file() enforces filename presence, extension allowlist, MIME allowlist, non-empty, size limit, and binary signature checks; raises 400/413 on failure.
- **Pagination implementation:** explicit page and page_size on GET /api/v1/reports; validates page>=1 and page_size within configured max.
- **Filtering/search implementation:** query-param filtering on reports (department_id, status, priority, issue_type) and system issues (status, category, reporter, search); dedicated search endpoints for reports.
- **Response model consistency/inconsistency:** mixed; many endpoints use response_model, several do not (raw dict/stream responses).
- **Error handling mechanism:** mostly per-endpoint HTTPException; no global custom 422 handler; custom global 429 handler exists.
- **HTTP 422 behavior:** default FastAPI validation error behavior (no custom override found).
- **HTTP 429 behavior:** custom JSON from app-level RateLimitExceeded handler: success=false, fixed message, error_code="RATE_LIMIT_EXCEEDED".
- **Health/readiness routes:** GET /health, GET /liveness, GET /readiness (readiness performs DB check with 503 on failure).

NON-BLOCKER

1. Singular/plural inconsistency in report paths (/api/v1/report vs /api/v1/reports/*) can still be documented as-implemented.
2. Mixed response model usage (typed + untyped responses) is documentable as-is.
3. Rate-limit strings are syntactically inconsistent (e.g., minute vs minutes) but still directly documentable from source.

